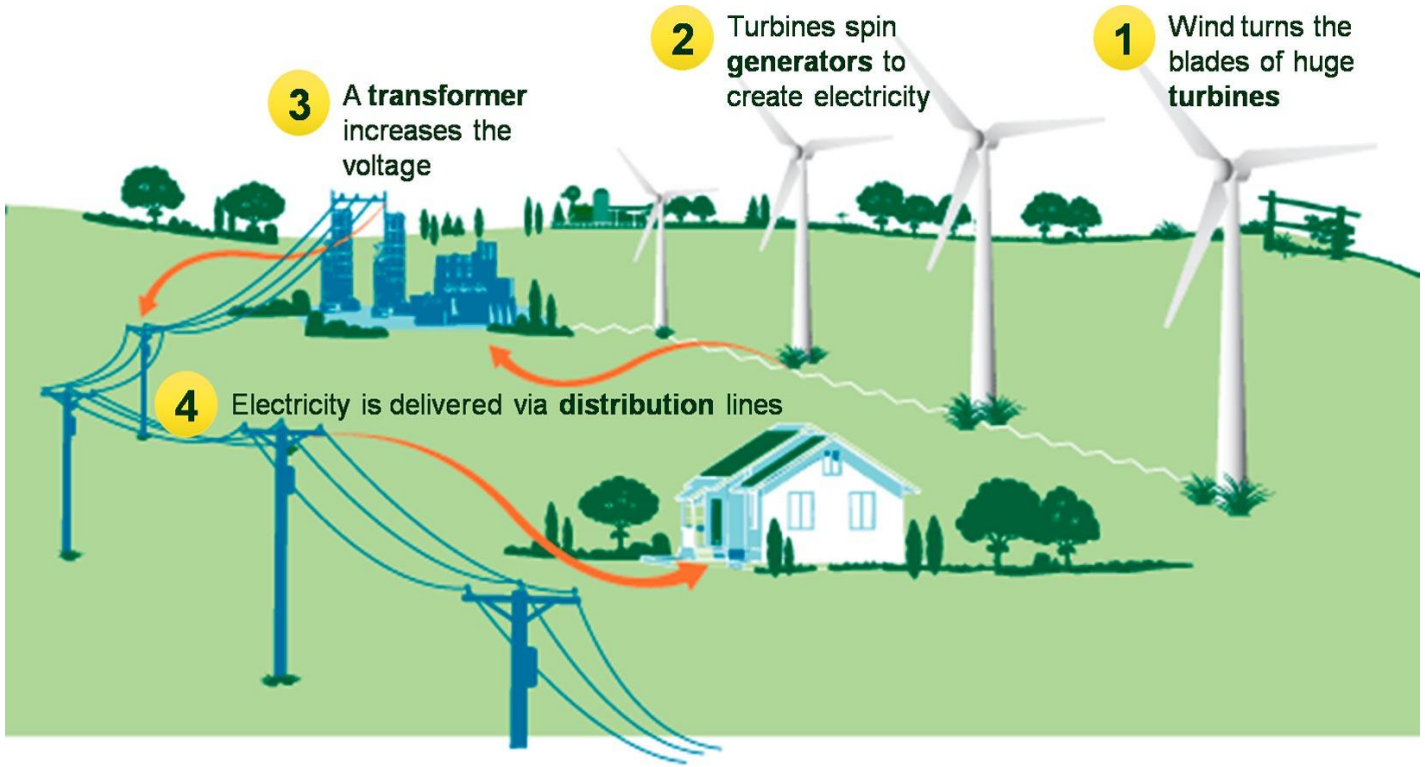
HAT CO. WINDMILL PROJECT

Helen Hoyos

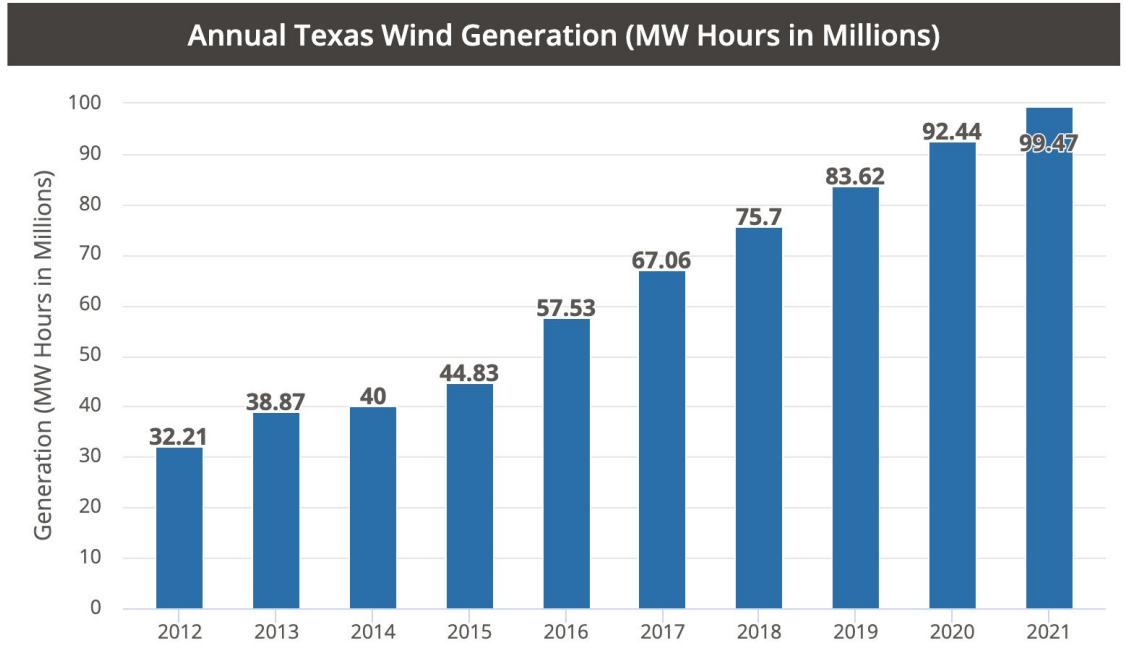
CASE STUDY FOR KEY ELEMENTS IN A PROJECT FINANCING

- 80 MW Wind Energy Turbine Project
- 4 Key Project Contracts
  - Lump sum turn-key EPC contract
  - Long term (25 yr) sale of electric energy to a utility/ISO
  - Operating company contract
  - Land Lease
- Key Financing Agreements
  - Project construction financing (3 year)
  - Project take out financing (15 year)
- Project financial analysis for developing an AT Levered IRR & NPV10
- Project sensitivity analysis

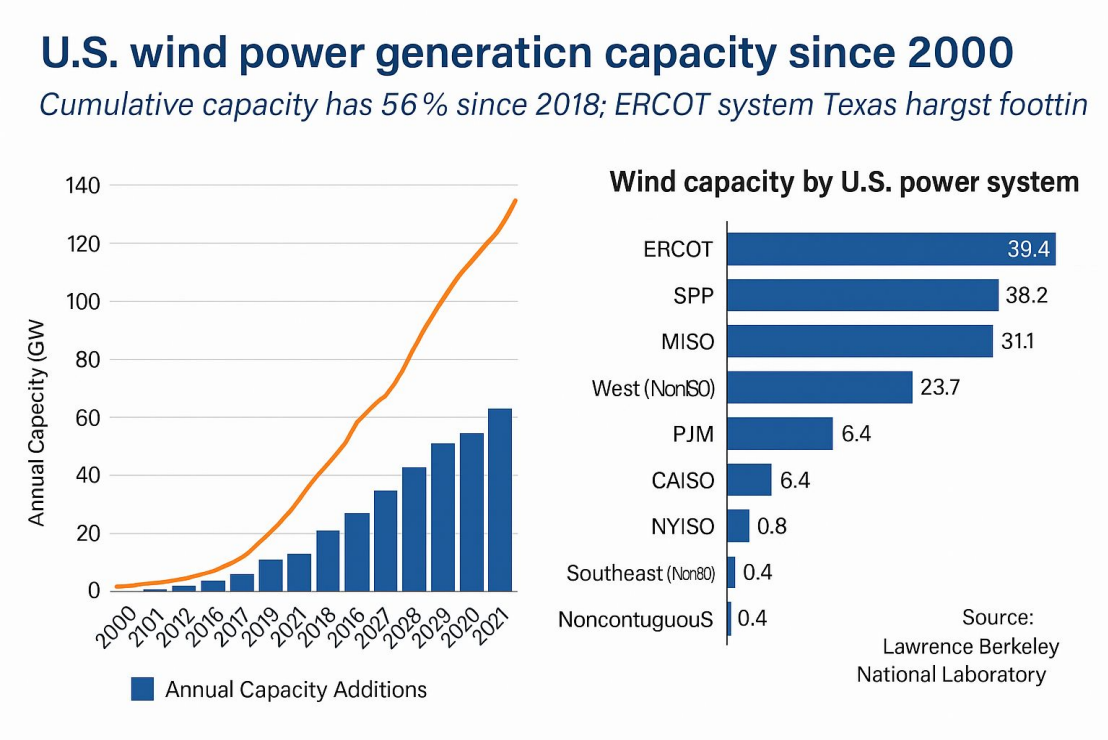
WIND TURBINE SYSTEM



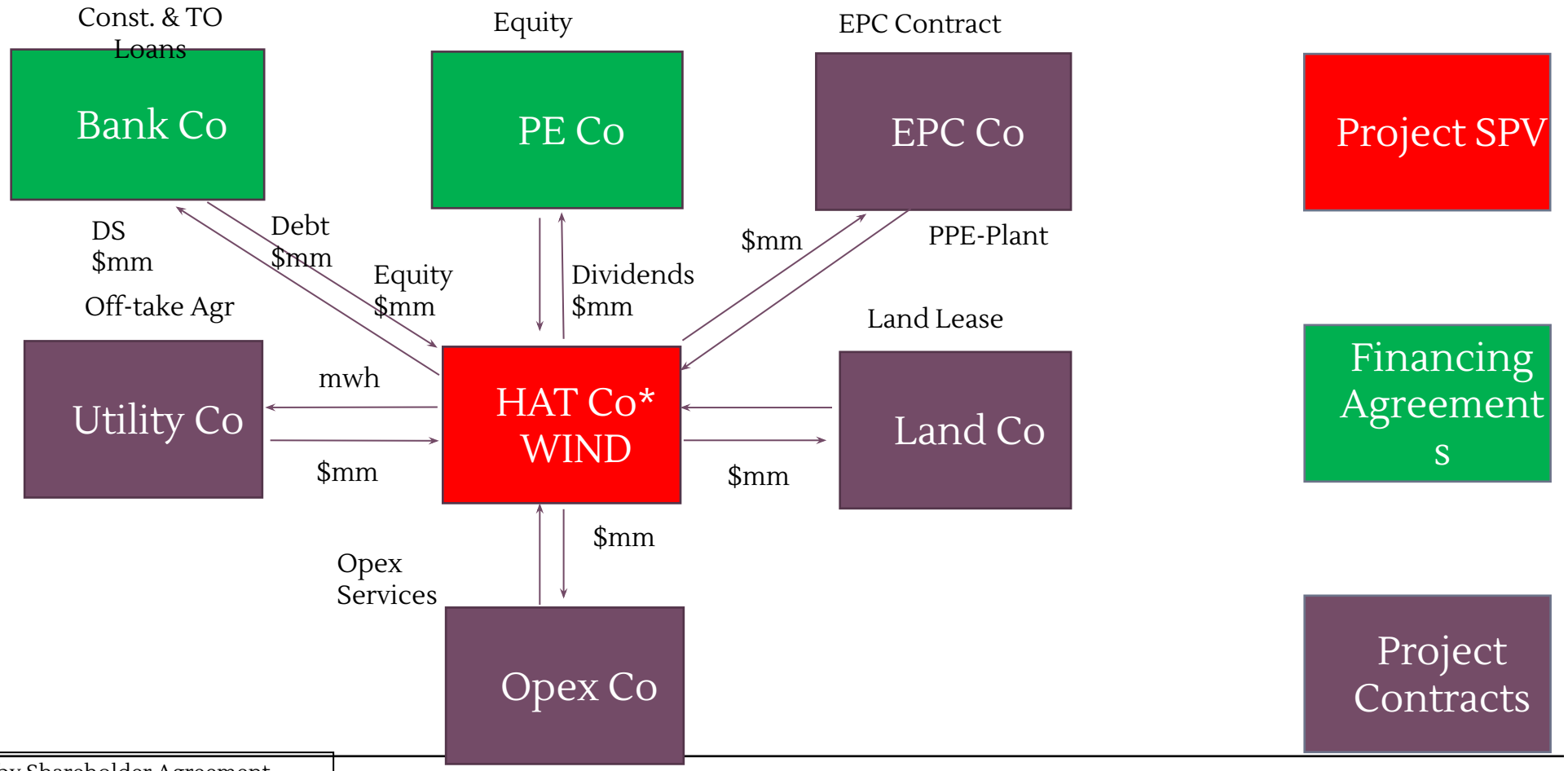
MARKET



Source: U.S. Energy Information Administration



SPV STRUCTURE FOR PROJECT FINANCING



* Governed by Shareholder Agreement

BASE DATA

- Technology: Utility-scale wind turbines
- Capacity: 80 MW
  - Capacity Factor: 40%
- Project Life: 25 years
- Fixed O&M: $42/kW/year
- Tax Rate: 21%
  - Tax Depreciation: 6 years
- No fuel or variable costs
- No carbon emissions

EPC TERM SHEET

- Buyer: HAT Co
- Seller: Engineering Co
- Contract Type: Lump-Sum Turn-Key
- Price: $126mm
- Scope: Definition of the Work
  - 80 MW Wind Energy Turbine
- Liquidated damages: Penalties for breach of contract
  - Schedule: 9.6mm
  - Performance: 4.9mm per 10% capacity loss
- Construction Schedule: 3 Years
  - Start Date: January 2025
  - Completion Date: December 2027
- Payment Schedule:
  - 10% Year 1: $12 mm
  - 40% Year 2: $48 mm
  - 50% Year 3: $60 mm
- Guarantees
  - Seller: Parent Guarantee by Engineering Co.
  - Buyer: Letter of Credit HAT Co.

SALE OF ELECTRICAL ENERGY TERM SHEET

- Buyer: Utility Co
- Seller: HAT Co
- Start date: 1 Jan 2028
- Term: 15 Years
- Volume:
  - 280,320 mwh per year
- Payment Mechanisms
  - Payments metered for units delivered i.e., kwh
  - Payments 30 days after invoice
  - Base price 2025 72.24 $/MWH
- Security
  - Parent guarantee by Utility Co
  - Letter of Credit
  - Reserves dedication pass through by HAT Co.
- Delivery Point
  - HAT Co on site sub-station
  - Delivered into Utility Co HV Transmission System

OPERATOR AGREEMENT TERM SHEET

- Buyer: HAT Co
- Seller: Opex Co
- Start date: COD of the plant, Start-up period
- Term: Economic life, Phase In Option
- Price:
  - Fixed Opex: $42/kw/yr 2025 with incentives @ 2.5%
- Security
  - Parent guarantee from HAT Co.
  - Letter of Credit from Opex Co.

Note: Land lease ($2/KW/yr @2.5%) is assumed to be covered for the SPV by the operator in the fixed opex payment

LAND LEASE TERM SHEET

- Lessor: HAT Co
- Lessee: Land Co.
- Annual Lease Payment: $2/kw/yr 2025 @ 2.5% escalation
- Term: 25 Years with option to renew
- Security: HAT Co. Guarantee

LAND TURBINE MAP

- Sunray, Tx
- 2,334 acres
- 20 4.2 mw turbines

INCOME STATEMENT

REVENUES

Income Statement		2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Project Year					1	2	3	4	5	6	7	8	9	10
Construction Year		1	2	3										
Revenues														
Production	MWH				280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320
Price	$/MWH	$72.24	$ 74.05	$ 75.90	$ 77.79	$ 79.74	$ 81.73	$ 83.78	$ 85.87	$ 88.02	$ 90.22	$ 92.47	$ 94.79	$ 97.15
Total Revenues	$mm				$ 21.8	$ 22.4	$ 22.9	$ 23.5	$ 24.1	$ 24.7	$ 25.3	$ 25.9	$ 26.6	$ 27.2

Income Statement	2038	2039	2040	2041	2039	2040	2041	2042	2040	2041	2042	2043	2041	2042	2043
Project Year	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Construction Year															
Revenues															
Production	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320	280,320
Price	$ 99.58	$ 102.07	$ 104.63	$ 107.24	$ 109.92	$ 112.67	$ 115.49	$ 118.37	$ 121.33	$ 124.37	$ 127.48	$ 130.66	$ 133.93	$ 137.28	$ 140.71
Total Revenues	$ 27.9	$ 28.6	$ 29.3	$ 30.1	$ 30.8	$ 31.6	$ 32.4	$ 33.2	$ 34.0	$ 34.9	$ 35.7	$ 36.6	$ 37.5	$ 38.5	$ 39.4

EXPENSES

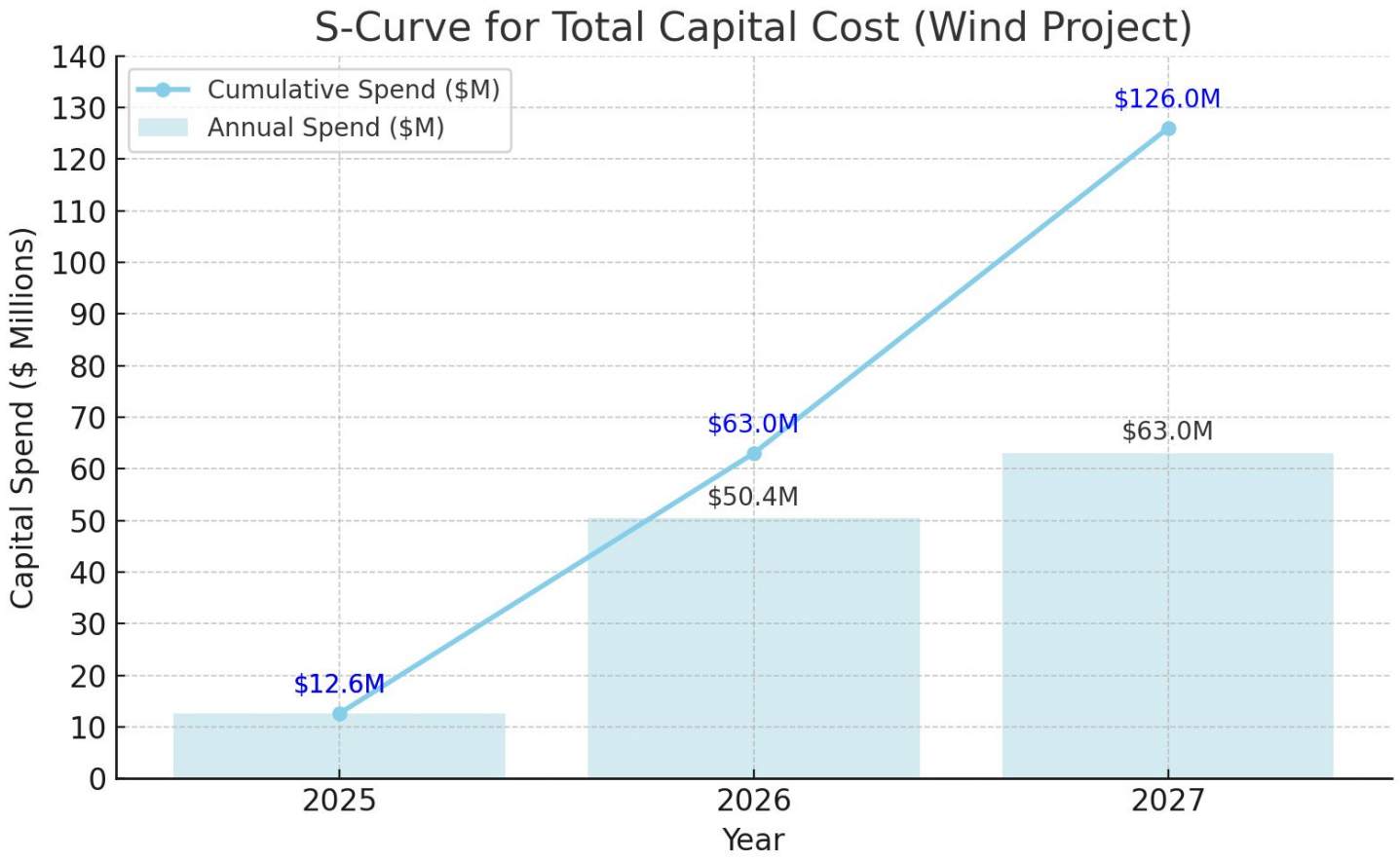
		2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
					1	2	3	4	5	6	7	8	9	10
Expense														
Fixed O&M	$/kw	42	$ 43.05	$ 44.13	$ 45.23	$ 46.36	$ 47.52	$ 48.71	$ 49.92	$ 51.17	$ 52.45	$ 53.76	$ 55.11	$ 56.49
Fixed O&M	$mm				$ 3.62	$ 3.71	$ 3.80	$ 3.90	$ 3.99	$ 4.09	$ 4.20	$ 4.30	$ 4.41	$ 4.52
Variable O&M	$/MWH	0	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -				
Variable O&M	$mm				$ -	$ -	$ -	$ -	$ -	$ -				
Heat Rate	btu/kwh				0	0	0	0	0	0				
Fuel Requirment	mmbtus				-	-	-	-	-	-				
Fuel price	$/mmbtu	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -				
Fuel Cost	$mm				$ -	$ -	$ -	$ -	$ -	$ -				
Total Expense	$mm				$ 3.6	$ 3.7	$ 3.8	$ 3.9	$ 4.0	$ 4.1	$ 4.2	$ 4.3	$ 4.4	$ 4.5

EBITDA

Project Year						1	2	3	4	5	6	7	8	9	10
Construction Year			1	2	3										
EBITDA	$mm					$ 18.2	$ 18.6	$ 19.1	$ 19.6	$ 20.1	$ 20.6	$ 21.1	$ 21.6	$ 22.2	$ 22.7
Start-up Opex Reserve					-1.8										
Pre-Tax Unlevered Cash Flow	$mm		-$12.6	-$50.4	-$64.8	$18.2	$18.6	$19.1	$19.6	$20.1	$20.6	$21.1	$21.6	$22.2	$22.7
Pre-Tax Unlevered IRR	%		14.79%												
NPV 10	$mm		$49.2												

Residual	7.5	-$12.6	-$50.4	-$64.8	$18.2	$18.6	$19.1	$19.6	$20.1	$20.6	$21.1	$21.6	$22.2	$193.1
Pre-Tax Unlevered IRR	%	15.26%												
NPV 10	$mm	$39.8												

CONSTRUCTION DEBT



CONSTRUCTION FINANCING

Construction Finance			Yr1	Yr2	Yr3
Constuction Debt BOY		$mm	$ -	$ 6.5	$ 32.6
Construction Debt Add		$mm	$ 6.3	$ 25.2	$ 31.5
Construction Debt EOY b4 IDC		$mm	$ 6.3	$ 31.7	$ 64.1
Construction Debt Ave		$mm	$ 3.2	$ 19.1	$ 48.4
Interest During Construction (IDC)		$mm	$ 0.2	$ 1.0	$ 2.4
Consturction Debt EOY a IDC		$mm	$ 6.5	$ 32.6	$ 66.5
			Yr1	Yr2	Yr3
Construction Equity per Year		$mm	$ 6.30	$ 25.20	$ 31.50
Construction Equity Total		$mm	$ 6.30	$ 31.50	$ 63.00
Final Construction Capital Structure			Yr1	Yr2	Yr3
Debt @ End of Construction Loan		$mm	$ 6.5	$ 32.6	$ 66.5
Equity @ End of Construction Loan		$mm	$ 6.30	$ 31.50	$ 63.00
Total Capital		$mm	$ 12.8	$ 64.1	$ 129.5

CONSTRUCTION LOAN AGREEMENT

Lender: Bank Co

Borrower: HAT Co

Term: 3 Years

Rate: 5% (Fixed)

Amortization:

- Amortized at Take-Out
- Includes Interest During Construction (IDC)

Equity: 50%

TAKE-OUT LOAN AGREEMENT

Lender: Bank Co

Borrower: HAT Co

Term: 15 Years

Rate: 6.5% (Fixed)

Amortization: Mortgage Style

Equity: 60%

Take-Out Capital Structure						
Take-Out Debt		$mm				$ 77.1
Equity EOY Construction Loan		$mm				$ 63.0
True-Up Equity		$mm				$ 14.1
Take Out Equity		$mm				$ 77.1
Total Take-Out Capital		$mm				$ 154.3

TAKE OUT FINANCE PROGRAM

Take-Out Loan Schedule		1	2	3	4	5	6	7	8	9	10
Debt Principal BOY	$mm	$ 77.14	$73.95	$70.55	$66.94	$63.08	$58.98	$ -	($4.65)	($9.61)	($14.89)
Take-Out Debt Principal	$mm	($3.19)	($3.40)	($3.62)	($3.85)	($4.10)	($4.37)	($4.65)	($4.96)	($5.28)	($5.62)
Take-Out Debt Interest	$mm	($5.01)	($4.81)	($4.59)	($4.35)	($4.10)	($3.83)	($3.55)	($3.25)	($2.92)	($2.58)
Take-Out Debt Service	$mm	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)
Debt Principal EOY	$mm	$73.95	$70.55	$66.94	$63.08	$58.98	$54.61	($4.65)	($9.61)	($14.89)	($20.51)
EBITDA	$mm	$ 18.2	$ 18.6	$ 19.1	$ 19.6	$ 20.1	$ 20.6	$ 21.1	$ 21.6	$ 22.2	$ 22.7
Take-Out Pre-Tax DSCR	.xx	2.22	2.27	2.33	2.39	2.45	2.51	2.57	2.64	2.70	2.77

PRE-TAX LEVERED CASH FLOW

Project Year						1	2	3	4	5	6	7	8	9	10
Consturction Year			1	2	3										
EBITDA	$mm					$ 18.2	$ 18.6	$ 19.1	$ 19.6	$ 20.1	$ 20.6	$ 21.1	$ 21.6	$ 22.2	$ 22.7
Debt service	$mm					($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)	($8.20)
Equity	$mm		$ (6.30)	$ (25.20)	$ (51.55)	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Pre-Tax Levered Cash Flow	$mm		$ (6.30)	$ (25.20)	$ (51.55)	$ 9.98	$ 10.44	$ 10.90	$ 11.38	$ 11.87	$ 12.37	$ 12.89	$ 13.42	$ 13.96	$ 14.51
Pre-Tax Levered IRR	%		6.52%												
Pre-Tax Levered NPV 10	$mm		($10.9)												

DEPRECIATION CALCULATION

Depreciation Calculations						
Gross Investment						
Take-Out Capital		$mm				$ 154
Additonal Equity Pre-COD		$mm				$ (23)
Total Gross Investment		$mm				$ 131

TAX DEPRECIATION

Depreciation Period		1	2	3	4	5	6
Gross Investment	$mm	$ 131	$ 131	$ 131	$ 131	$ 131	$ 131
Depreciation Rate	$mm	20.00%	32.00%	19.20%	11.52%	11.52%	5.76%
Tax Depreciation	$mm	$ 26.19	$ 41.91	$ 25.15	$ 15.09	$ 15.09	$ 7.54
Cumulative Depreciation		$ 26.19	$ 68.11	$ 93.25	$ 108.34	$ 123.43	$ 130.97
Net Investment	$mm	$ 105	$ 63	$ 38	$ 23	$ 8	$ -

AFTER TAX INCOME

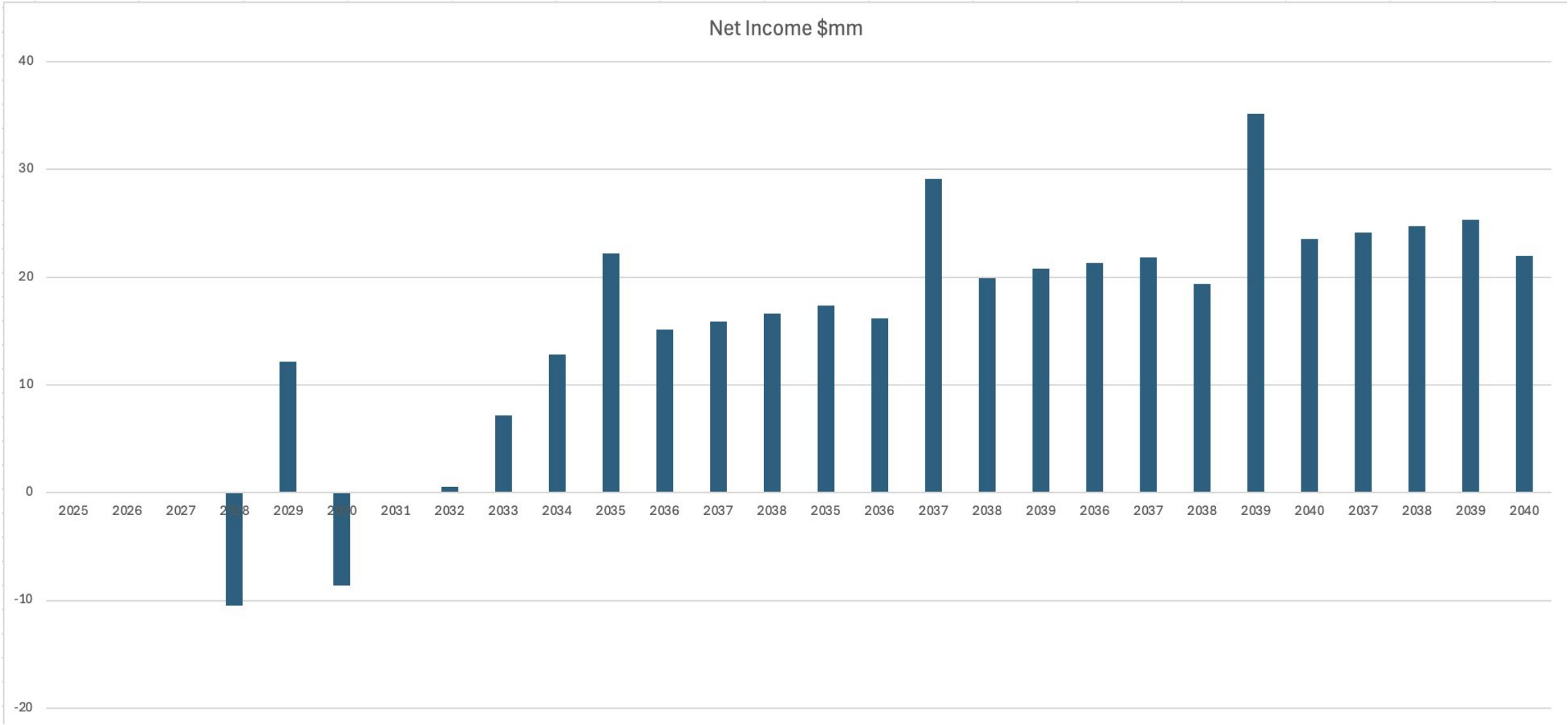
After-Tax Income			2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Project Year						1	2	3	4	5	6	7			
Construction Year			1	2	3										
EBITDA	$mm					$ 18.19	$ 18.64	$ 19.11	$ 19.59	$ 20.08	$ 20.58	$ 21.09	$ 21.62	$ 22.16	$ 22.71
Interest Expense	$mm					$ (5.01)	$ (4.81)	$ (4.59)	$ (4.35)	$ (4.10)	$ (3.83)	$ (3.55)	$ (3.25)	$ (2.92)	$ (2.58)
Depreciation	$mm					$ (26.19)	$ (41.91)	$ (25.15)	$ (15.09)	$ (15.09)	$ (7.54)	$ -	$ -	$ -	$ -
EBT	$mm					$ (13.02)	$ (28.08)	$ (10.62)	$ 0.15	$ 0.89	$ 9.20	$ 17.54	$ 18.37	$ 19.24	$ 20.13
Federal Tax	$mm					$ 2.73	$ 5.90	$ 2.23	$ (0.03)	$ (0.19)	$ (1.93)	$ (3.68)	$ (3.86)	$ (4.04)	$ (4.23)
Tax Credit	$mm					$ -	$ 46.28	$ -	$ -	$ -	$ -	$ 1.00	$ -	$ -	$ -
Federal Tax a/ Credit	$mm					$ 2.73	$ (40.39)	$ 2.23	$ (0.03)	$ (0.19)	$ (1.93)	$ (4.68)	$ (3.86)	$ (4.04)	$ (4.23)
Net Income	$mm					$ (10.29)	$ 12.31	$ (8.39)	$ 0.12	$ 0.70	$ 7.27	$ 12.86	$ 22.23	$ 15.20	$ 15.90

AFTER-TAX DEBT SERVICE COVER RATIOS

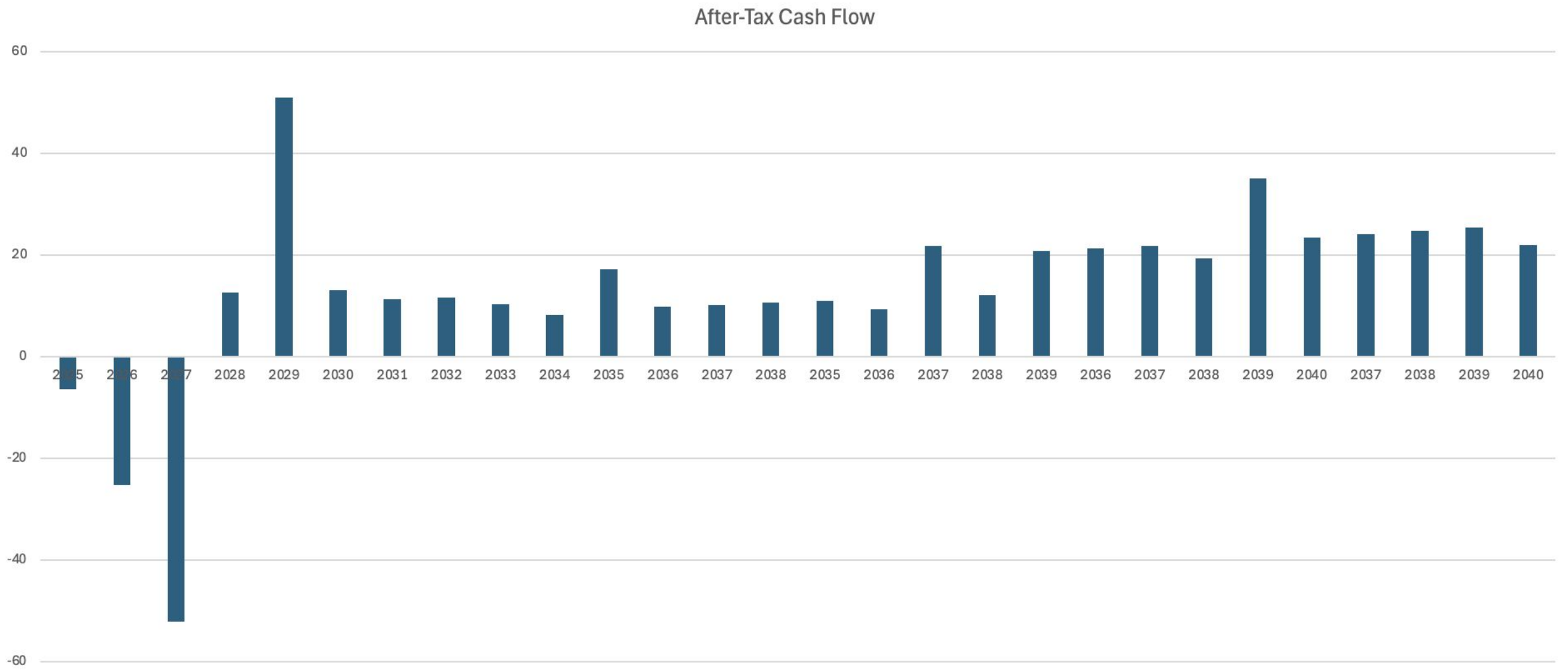
- DSCR = Cash Flow Available for Debt Service in a Year / Debt Service in a Year

- Estimated = 2.8
- Minimum = 2.0

NET INCOME


Net Income $mm
40
30
20
10
0
-10
-20
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
2036
2037
2038
2035
2036
2037
2038
2039
2036
2037
2038
2039
2040
2037
2038
2039
2040

AFTER-TAX CASH FLOW



OTHER ASPECTS IN PROJECT ANALYSIS

- Sensitivity Analysis
- Liquidated Damages
- Developer Returns
- GHG Analysis
- Capital Raised Investment Term Sheet

PROJECT SENSITIVITY ANALYSIS:

This analysis evaluates how changes in power price, capacity factor, CapEx, and other key variables affect the project's financial performance.

Variable	Low Case	Base Case	High Case	IRR Impact
Power Price ($/MWh)	$65.00	$72.24	$80.00	12.4% / 16.8%
Capacity Factor	36%	40%	44%	12.1% / 16.3%
O&M Costs ($/kW/yr)	$50	$42	$35	13.5% / 15.9%
CapEx ($ million)	$165	$155.40	$145	13.3% / 16.2%
Debt Rate (%)	6.50%	5.00%	3.50%	13.2% / 16.0%
PPA Term (years)	15	20	25	13.4% / 15.7%

LIQUIDATED DAMAGES

SCHEDULE

- Loss per year: 9.6mm
- Loss per month: 0.8mm
- Loss per day: 26339.44

PERFORMANCE

- Performance Loss per 10% of 80 MW: 4.9mm
- Performance Loss per 1% of 80 MW: 0.49mm

DEVELOPER RETURNS

Development Fee:

- Fee Rate: 5% of Total Project CapEx
- Total CapEx: $68 million (for 80 MW project)
- Development Fee Amount: $3.4 million

Fee Stru

Milestone	Portion of Fee Paid
Financial Close	50%
Start of Construction (NTP)	25%
Commercial Operation Date	25%

EMISSIONS COMPARISON

Wind energy is one of the cleanest sources of electricity, with **near-zero lifecycle greenhouse gas emissions**. Here's how it compares to traditional energy sources:

Here's how it compares to traditional energy sources:

Energy Source	CO_2 Emissions (g/kWh)
Wind (Onshore)	~11 g CO_2/kWh
Natural Gas	~450 g CO_2/kWh
Coal	~900 g CO_2/kWh

EMISSIONS COMPARISON (CONTINUED)

ESG Impact Highlights

- Equivalent to taking **100,000+ cars** off the road annually
- Aligned with **UN SDG 7 & 13** : Affordable Clean Energy & Climate Action
- Enhances investor ESG ratings and compliance with green finance frameworks

Annual Emissions Avoided

Project Capacity: 80 MW

Annual Generation: ~280,000 MWh

Compared to Natural Gas: ~122,000 metric tons of CO_2 avoided per year

Compared to Coal: ~252,000 metric tons of CO_2 avoided per year

CAPITAL RAISE INVESTMENT TERM SHEET

- Security: 1000 Common Shares of HAT Co
- Seller: Founders of HAT Co
- Funders: Private Equity Co Consortium
- Shareholder Agreement
  - Capital structure: 50% Equity/50% Debt
  - Amount of equity being raised: $126mm
  - Equity contribution per share: $500,880/Share Par Value
  - Paid in to fund the S Curve
  - Dividends: Paid Annually
  - Development Fee of 5% on Capex Paid to Seller at NTP
- Standard Representations and Warranties
- Governance Provisions
  - Board Seats to Common Shareholder
  - Voting Rules: Decision by Majority Vote of Shareholders
- Closing on 1 January 2025
- Standard Confidentiality Provisions
- Each party pays their own legal & advisory fees

INVESTMENT HIGHLIGHTS

Critical Electric Power Infrastructure for the US economy

Low-cost base-load electricity

Reliable in terms of availability at 90%

50% debt financed with long-term loan

Excellent equity return with IRR at 15%

At levered NPV 10% of $53mm

Almost no carbon emissions

30% tax credit